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APPLIED RESEARCH

AI-Driven Post-Earthquake Emergency Material Demand Prediction: Integrating RAG With Reasoning Large Language Model

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ABSTRACT The supply of emergency rescue materials plays a pivotal role in post-earthquake relief operations. However, varying disaster scenarios generate differentiated demands for emergency resources, where factors such as seasonal impacts, geographic environment of affected areas, population density, and requirements of emergency response protocols significantly influence the categories and quantities of required supplies. Current research predominantly focuses on essential materials, while determinations of scenario-specific material demands rely heavily on expert empirical inference, resulting in discrepancies between model predictions and actual requirements. This study proposes a reasoning-enhanced large language model (LLM) framework integrated with Retrieval-Augmented Generation (RAG) technology for post-earthquake emergency material demand prediction. By simulating expert decision-making processes, we construct an emergency knowledge base amalgamating standardized protocols, historical seismic case data, and disaster scenario characteristics extracted from digital resources. The framework employs RAG to enhance domain-specific knowledge integration within the reasoning model, utilizing Chain-of-Thought generation to produce differentiated prediction schemes that specify material categories and per-capita demand metrics. Through dynamic updating of population statistics and material requirements via post-disaster network information monitoring, the system achieves real-time demand prediction based on evolving victim counts and individualized allocation parameters. Validation through expert evaluations using the 2013 Ya'an Lushan earthquake and simulated disaster scenarios demonstrates effectiveness, with successful practical implementation observed in the January 7, 2025, Dingri M6.8 earthquake case.

INDEX TERMS Emergency relief supply prediction, large language models, retrieval-augmented generation.

I. INTRODUCTION

Earthquakes, as sudden and highly destructive natural disasters, are extremely difficult to accurately predict in advance. They often occur abruptly, characterized by uncertain timing and unpredictable severity, posing significant risks to human lives and property. Once a destructive earthquake occurs,

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emergency rescue operations must be initiated immediately. An efficient emergency relief supply system serves as the backbone of these efforts, directly influencing the speed and effectiveness of disaster response. Effective prediction of relief material demand is therefore crucial—it lays the foundation for material stockpiling, procurement planning, transportation coordination, and equitable distribution.

In recent years, researchers worldwide have extensively explored methods for predicting disaster relief material

needs. Given that post-earthquake demands are sudden, hard to estimate, time-sensitive, phase-dependent, regionally varied, and seasonally influenced, selecting an appropriate and effective prediction method is essential. Such methods help avoid shortages or surpluses in supply, as well as delays in distribution.

With the widespread adoption of internet technologies, social media and other online platforms have become vital channels for public communication during crises. Incidents of affected communities using shared documents to seek help are no longer isolated cases. By analyzing content from social media and online forums, real-time updates can be extracted to rapidly assess needs in disaster-stricken areas. Collecting survivor requests and aid appeals allows for differentiated demand predictions, enabling precision in resource allocation. This approach prioritizes severely affected regions and vulnerable groups, minimizing waste and inequitable distribution. Understanding demand quantities and types also helps relief agencies optimize procurement, storage, and transportation, manage supply chains efficiently, and mobilize resources swiftly, reducing disaster-induced losses. Effective demand forecasting methods thus support decision-makers in devising scientifically sound rescue plans, improving timeliness and accuracy.

Traditional prediction methods primarily rely on seismic parameters like magnitude, depth, and location, with some studies incorporating demographics, transportation, and weather conditions. However, disaster scenarios vary widely, and conventional approaches, which input fixed parameters to predict common supplies (e.g., food, water, tents), struggle to account for complex influencing factors. For instance, earthquakes in mountainous winter regions necessitate additional cold-weather gear, while prolonged post-disaster rain calls for waterproof tarps and drainage equipment.

This study employs large language models to automatically extract multidimensional features from online sources—including news reports, government bulletins, and social media—such as timing, aftershocks, terrain, weather, socio-economic conditions, demographics, medical resource distribution, and transportation status. To alleviate the phenomenon of LLMS creating illusions due to the lack of domain expertise, the emergency knowledge base was used to apply retrieval-augmented generation (RAG), which improved reliability and interpretability. The model first selects standardized supplies (e.g., tents, medicine, lighting equipment) from the knowledge base, then combines them with disaster data to generate context-aware predictions of material types and per-capita demand.

To address varying post-disaster information dynamics, a phased prediction strategy is proposed. During the early “blackout period,” a BP neural network rapidly estimates casualties based on population density and seismic parameters. Simultaneously, the LLM analyzes real-time online data—hospital reports, social media pleas—to refine casualty figures. Combining these with previously predicted

per-capita demands yields total material requirements. This hybrid approach resolves early-stage data scarcity issues while dynamically adjusting predictions to reflect evolving disaster conditions, ensuring both timeliness and accuracy.

II. RELATED WORKS

Prior to 2008, there were relatively few researchers in China focusing on the issue of emergency relief supply prediction, and the predominant methodology involved constructing multiple regression models. For instance, researchers such as Gao-Zhong Nie [1] from the China Earthquake Administration cite utilized data from the “Integrated Natural Disaster Information Sharing” website to develop demand estimation formulas for over 80 types of emergency supplies required after earthquakes. These formulas were designed to enable rapid post-disaster demand predictions.

However, the catastrophic Wenchuan Earthquake in May 2008 brought widespread attention to the challenge of ensuring adequate relief supply distribution to support disaster response operations. As a result, the problem of accurately forecasting the types and quantities of materials needed in disaster zones became a key concern in emergency management.

Since then, numerous methods for emergency relief demand prediction have been developed both domestically and internationally, including—but not limited to—case-based reasoning, grey forecasting, regression analysis, and neural networks. Additionally, an increasing number of scholars have turned their attention to dynamic demand forecasting across different post-disaster phases, recognizing the evolving nature of relief needs over time.

A. POST-EARTHQUAKE EMERGENCY RELIEF DEMAND PREDICTION METHODS

The principle of case-based reasoning is to infer possible current situations based on past cases and predict the required materials by understanding the material situation of similar disaster cases [2]. To use this method, a historical earthquake case set needs to be established first. Usually, the selected earthquake disaster characteristics include magnitude, population density, epicentral distance, and focal depth, to quickly find the most similar situation to the target situation for prediction [3]. The disadvantage of the case-based reasoning method is that it is difficult to collect typical and effective cases to deal with sudden destructive earthquakes. The multiple linear regression analysis method finds the changing patterns of the influence of multiple earthquake disaster characteristics on material demand based on multiple sets of statistical data, and then predicts the emergency material demand after obtaining the relationship formula. Bin et al. [4] developed a typhoon emergency material demand prediction model based on fine grid geographic data. Ziyang and Bin [5] combined the earthquake risk assessment method with the deterministic method to analyze the emergency material demand prediction. Bin et al. [6] proposed an emergency

material demand prediction model based on prospect theory considering the decision-maker's perceived value. From the perspective of linear programming, Park et al. [7] purchased and distributed perishable humanitarian materials to the demand areas with a limited budget under the condition of uncertain demand, with the goal of minimizing the expected total shortage. The multiple linear regression method can obtain good data interpretability, but it faces the problems of difficulty in establishing formulas and low applicability of the formulas. The grey model prediction method (grey models, GM) is a method used for short-term prediction when the data volume is small and the direct correlation of the data is not clear. Usually, the GM(1,1) model or GM(2,1) model is used to predict the problem. Geng and Hou [8] established a grey Markov chain model based on geological, environmental and human factors to predict the number of people affected by earthquakes after the disaster, and then estimated the material demand; Zhang et al. [9], [10] established a grey discrete Verhulst model to predict the number of injured people and medicines in the earthquake-stricken area; Lili et al. [11] and Hu et al. [12] respectively used the improved metabolism GM(1,1) model to predict the emergency materials needed after earthquakes and floods; Chen and Liu [13] transformed the material demand problem into predicting the mortality and survival rates of the disaster area population, and established a GM(3,0) model based on three factors: mortality rate, house collapse rate and earthquake intensity level. The artificial neural network system (artificial neural networks, ANN) originated in the 1940s and is a data processing model constructed by simplifying and simulating the structure and processing process of the biological nervous system [14], which is the origin and foundation of the artificial intelligence boom. In addition to artificial neural networks, traditional machine learning methods have also been applied to the prediction of emergency material demand after disasters. For example, Wang et al. [15] proposed a support vector machine (SVM) model using principal component analysis (PCA) and Bayesian optimization (BO) to predict the number of casualties and material demand in earthquakes. Currently, the neural network algorithms mainly used in the research of emergency material demand prediction at home and abroad are BP (backpropagation) neural networks. For example, Zhangwei and Yunxia [16] used BP neural networks to predict the quantity of emergency materials needed for power networks, and Fenglin and Jian [17] used BP neural networks to predict the number of casualties in earthquakes. To address the problem that neural networks are prone to falling into local optimal solutions, other methods can be introduced to optimize the parameters of neural networks. Xiang et al. [18] introduced the clonal selection particle swarm algorithm on the basis of the ordinary BP algorithm to establish a power demand prediction model. In addition, other optimization algorithms include genetic algorithm (GA) [19], GKNN (grey K-nearest neighbor, GKNN) algorithm, principal component

analysis, ant colony optimization (ACO), grey relational analysis, additional momentum and adaptive adjustment of learning rate, forest optimization algorithm, safety stock model, metabolism method, etc. [20] Optimization algorithms such as GWO-BP [21] and IACO-BP [22] have also been used for post-disaster material prediction. Applications of Large Language Models in Prediction Tasks and Disaster Emergency Response Fields

B. APPLICATIONS OF LARGE LANGUAGE MODELS IN PREDICTION TASKS AND DISASTER EMERGENCY RESPONSE FIELDS

Large Language Models (LLMs), often referred to as LLMs for short, possess powerful context understanding and generation capabilities. They have achieved breakthroughs in various tasks within the field of natural language processing. In recent years, their application scope has gradually expanded to predictive tasks, such as forecasting trends in financial markets [23], assisting in medical diagnoses, and predicting social events. By capturing temporal features in text, LLMs have been applied to tasks such as wind speed prediction [24] and stock price prediction [25]. Research shows that LLMs can effectively integrate historical data and external knowledge to enhance the stability of long-term predictions. Moreover, LLMs excel in handling multimodal information such as text, images, and sensor data. Utilizing large models to process multimodal data and integrate multi-source information for predicting and evaluating complex scenarios has become a new direction. Schoenegger et al. [26] found that the accuracy of group predictions made by LLMs can rival that of human experts. Consequently, LLMs have gradually been applied in the emergency domain for tasks such as disaster prediction and rapid disaster assessment. Mousavi et al. [27] utilized LLMs to analyze social media text and images to estimate earthquake intensity. In complex scenarios, LLMs can generate high-precision prediction results through logical reasoning and context learning. Wang et al. [28] proposed a real-time framework for estimating earthquake casualties based on LLMs, achieving rapid response within hours after a disaster by analyzing social media and news data. Han et al. [29] used LLMs to improve the analysis of social media text and enhance the accuracy of earthquake impact assessment. Biswas et al. [30] developed an AI framework that leverages LLMs to process multi-source heterogeneous data after earthquakes and predict rescue needs in the Turkish earthquake. LLMs have also been employed to generate emergency plans and optimize resource allocation. For instance, Xia et al. [31] designed an AI-driven medical supply scheduling system to enhance the efficiency of public health event responses. Mostajabdaveh et al. [32] proposed a branch-and-price algorithm that combines the reasoning capabilities of LLMs to achieve fair and efficient material distribution.

C. APPLICATION OF RETRIEVAL-AUGMENTED GENERATION IN DISASTER EMERGENCY RESPONSE

Retrieval-augmented generation enhances the dynamic adaptability and accuracy of large language models in specialized fields by integrating external knowledge bases. Compared with traditional methods, using LSTM models requires a large amount of training data, has high model update costs, and is difficult to incorporate emerging rules and events in a timely manner. Its immediate learning ability for new input data is usually weak. Using the GM(1,1) model for prediction, its structure is fixed and has almost no adaptability to system dynamic changes. Each time there is significant new data, it usually needs to be re-modeled. With RAG, there is no need for re-training. When new knowledge, new rules, or unexpected events occur, only the associated vector database needs to be updated to retrieve the latest relevant information in real time and feed it to the LLM. Chen et al. [33] combined disaster knowledge graphs with LLMs to provide structured knowledge support for emergency decision-making. Sufi [34] systematically reviewed the application of LLMs in open-source disaster intelligence, emphasizing that knowledge retrieval is crucial for the credibility of generated results. RAG compensates for the limitations of LLMs in domain-specific knowledge by dynamically retrieving specialized literature, historical cases, and real-time data. For instance, Lei et al. [35] proposed an RAG-based disaster management framework to support multilingual and multimodal disaster analysis and demand prediction. These studies demonstrate that incorporating authoritative data sources—such as seismic intensity databases and emergency policy texts—can effectively reduce the likelihood of LLMs generating erroneous information and mitigate the occurrence of hallucinations.

III. METHODOLOGY

A. RETRIEVAL-AUGMENTED GENERATION WITH EMERGENCY KNOWLEDGE BASES

The emergency knowledge base is constructed by collecting expert knowledge and online information, employing retrieval-augmented generation (RAG) to mitigate the lack of specialized knowledge and reduce hallucinations in large language model outputs. Chain-of-thought prompting enhances the reasoning capabilities of the model, enabling comprehensive analysis and logical deduction of disaster-related information. This facilitates differentiated predictions under varying disaster conditions, ultimately yielding accurate forecasts of required relief supplies and their quantities.

1) INFORMATION COLLECTION

This study systematically establishes a multi-dimensional data support framework for earthquake emergency management, integrating legal regulations, historical cases, and contingency plans into a tripartite research paradigm.

Regulatory Framework: Through official channels including the Ministry of Emergency Management and the National Development and Reform Commission, we compiled 12 specialized earthquake and emergency supply regulations and 23 disaster-related policies. Notably, the Key Material Classification Directory for Emergency Support categorizes 65 major and 218 sub-classes of supplies—including tents, medical equipment, and generators—providing a standardized baseline for demand forecasting.

Historical Case Database: The repository encompasses metadata from 216 earthquakes above magnitude 5 (2012–2025). For 38 major earthquakes exceeding magnitude 6 (e.g., the 2022 Luding M6.8 and 2023 Jishishan M6.2 events), distributed web crawlers extracted 65,000+ multi-source records from government bulletins, news reports, and academic platforms. After deduplication, spatiotemporal alignment, and NLP processing, 12,000+ structured datasets were generated, covering 12 dimensions such as seismic damage patterns and resource consumption.

Contingency Plan System: Breaking from fragmented traditional approaches, 35 hierarchical plans were integrated—including the National Earthquake Emergency Plan, 31 provincial plans, and localized protocols for high-risk areas. This formed a three-tier response framework (“national-provincial-local”).

By synergizing regulatory compliance, machine-learning-ready case data, and institutional logic from contingency plans, this work achieves systematic unification of policy norms, empirical evidence, and operational mechanisms.

2) KNOWLEDGE RETRIEVAL

Document indexing is the core component in RAG (Retrieval-Augmented Generation) technology that bridges raw data and efficient retrieval. It involves segmenting parsed documents into short knowledge chunks (chunks) and constructing vector indices. The primary objective is to transform structured documents into quickly retrievable knowledge units while establishing a multi-dimensional semantic association network. For complex queries, answers are often scattered across multiple text chunks, as isolated chunks cannot provide complete context. Common text segmentation strategies involve splitting documents into fixed token sizes (e.g., 100, 256, or 512 tokens). Larger chunks offer richer context but may introduce noise and higher processing costs, whereas smaller chunks reduce noise at the expense of insufficient context.

Vector embedding serves as the core module for semantic understanding in RAG technology, mapping unstructured text into a high-dimensional vector space to establish semantic relationships between documents. This enables subsequent similarity calculations. The model is designed to enhance comprehension of inter-document relationships, improving the accuracy of knowledge retrieval. In a typical RAG framework, embedding models include a sparse encoder and a dense retriever, each providing distinct semantic representations. The sparse encoder relies on word frequency

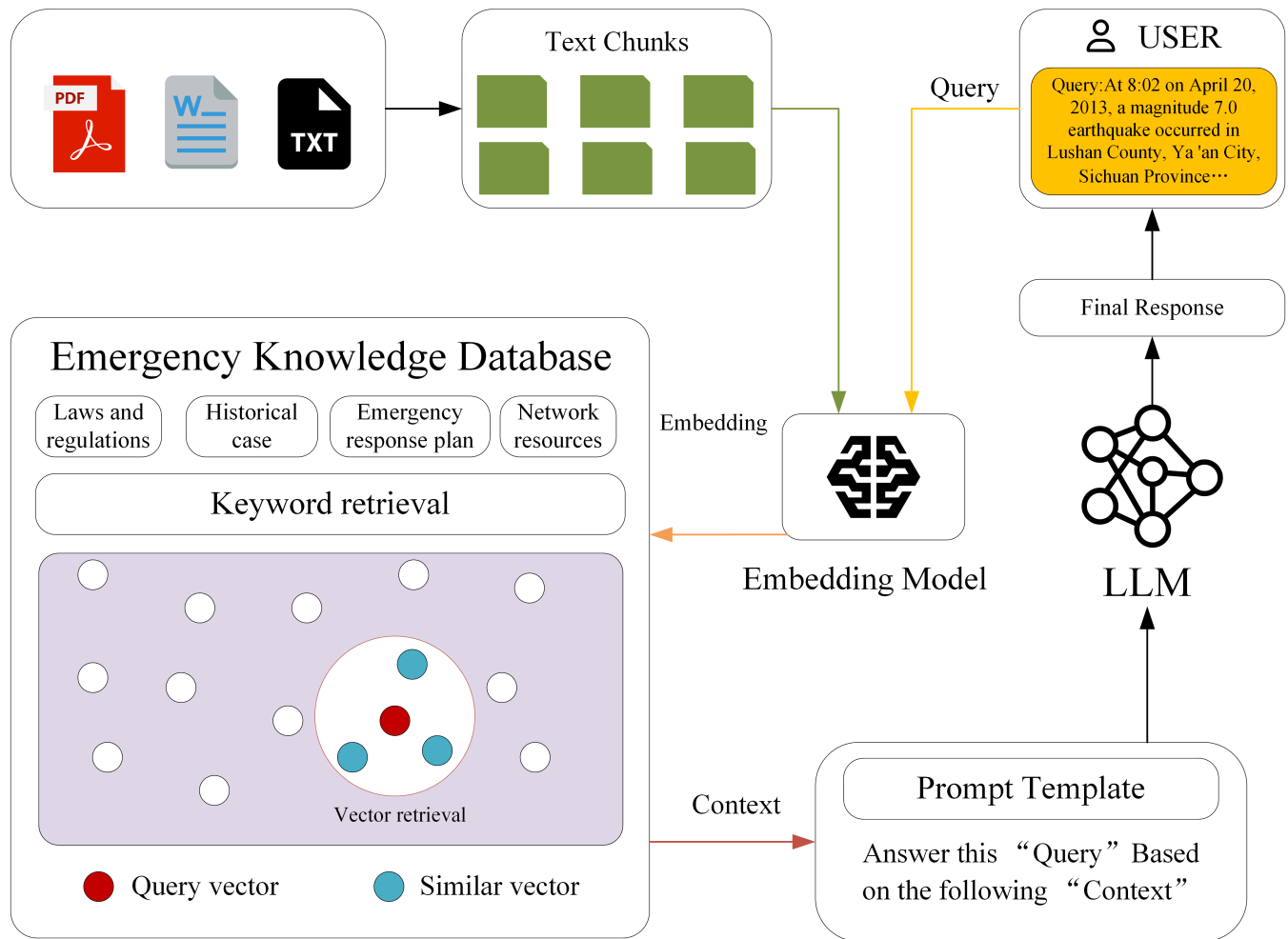


FIGURE 1. RAG flow chart.

statistics, while the dense retriever leverages deep learning models for finer-grained semantic encoding.

MTEB (Massive Text Embedding Benchmark) is the most authoritative benchmark for evaluating text embedding models, developed by Hugging Face. By querying the MTEB leaderboard and filtering open-source models compatible with the Chinese language, we assessed performance rankings and ultimately selected the gme-Qwen2-VL-7B-Instruct model as our embedding solution.

Hybrid retrieval serves as a pivotal technology for enhancing Retrieval-Augmented Generation systems, dynamically balancing recall and precision by synergizing the complementary strengths of keyword matching and vector-based semantic retrieval. In earthquake emergency supply demand forecasting, this hybrid approach achieves an optimal equilibrium between specificity and generalizability.

The keyword retrieval component rapidly identifies material classifications within regulatory standards, ensuring precise recall of critical supplies—particularly during the 0-72-hour golden window post-disaster—by prioritizing items like hemostatic agents and life detection equipment. Meanwhile, vector retrieval transcends literal matching,

capturing semantically linked demands from social media and historical cases: for instance, mapping “insufficient temporary shelters” to prefabricated housing needs, or inferring oxygen concentrator requirements from “plateau hypoxia,” thereby addressing supply gaps undefined by conventional regulations.

A distinguishing advantage lies in its dynamic scenario adaptation. The system employs a weighting mechanism that prioritizes keyword retrieval during initial rescue phases while progressively increasing vector retrieval weightage for reconstruction, effectively harmonizing institutional protocols with operational experience.

3) AUGMENTED GENERATION

Leveraging LLMs to generate the final response requires the integration of retrieved contextual information to produce coherent and precise textual answers. Selecting the most relevant retrieved segments enhances the accuracy of generated responses. By fully utilizing retrieved knowledge while minimizing computational resource consumption and maximizing the relevance-knowledge similarity,

TABLE 1. Massive Text embedding benchmark.

Model	Data (MB)	Dim	Avg	Class	Cluster	Pair	Reorder	Retrieval	Similarity
gme-Qwen2-VL-7B-Instruct	31629	3584	70.72	78.13	61.72	75.85	69.61	75.38	63.34
gme-Qwen2-VL-2B-Instruct	8427	1536	68.02	75.90	57.73	73.45	67.17	72.38	61.09
e5-mistral-7b-instruct	13563	4096	59.92	72.96	52.30	66.31	61.38	61.75	48.34
multilingual-e5-large	2136	1024	58.08	69.80	48.23	64.52	57.45	63.65	45.81
multilingual-e5-base	1061	768	55.58	67.71	40.68	63.45	57.04	61.59	43.75

reasoning-capable large language models can effectively improve interpretability and mitigate issues like domain knowledge gaps and hallucinations. We will elaborate on this approach in detail in the following subsection.

B. REASONING TO GENERATE PREDICTIONS

1) PREDICTION OF THE NUMBER OF AFFECTED PEOPLE IN THE EARLY POST-EARTHQUAKE PERIOD

Following an earthquake, the “information blackout period” immediately after the event often makes it difficult to obtain sufficient seismic data to construct multi-feature neural networks. Additionally, due to the scarcity of historical earthquake damage datasets, deep learning methods tend to perform poorly. Therefore, this study adopts a simpler neural network structure—specifically, a backpropagation (BP) neural network—to predict casualty figures.

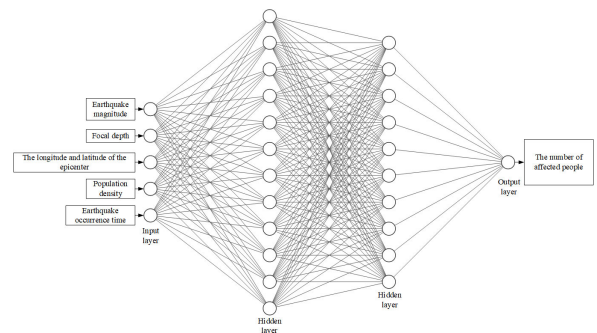
Given the lack of comprehensive disaster data during the early stages post-earthquake, the model utilizes five easily obtainable yet strongly correlated features: earthquake magnitude, focal depth, epicenter coordinates (latitude/longitude), population density at the epicenter, and occurrence time.

To optimize the BP neural network’s performance, the whale optimization algorithm (WOA) is employed for parameter tuning. The final architecture consists of:

- Input layer: 5 nodes, corresponding to the 5 features
- Hidden layers: 2 layers with 12 and 10 nodes, respectively
- Output layer: 1 node (predicted casualty count)

In addition to the BP neural network, methods such as LSTM and the improved GM(1, 1) model are also commonly employed for post-disaster casualty prediction. We compare these approaches with the BP neural network and assess their predictive performance using error metrics, as defined below. The formula for Mean Squared Error (MSE) is:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2 \quad (1)$$

**FIGURE 2.** BP neural-network.

The formula for Root Mean Squared Error (RMSE) is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2} \quad (2)$$

The formula for Mean Absolute Error (MAE) is:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - x_i| \quad (3)$$

The predicted value is y_i , the actual value is x_i , and n is the number of samples.

For comparative analysis, we conduct controlled experiments using an improved LSTM model and the GM(1, 1) model, alongside our proposed method. Each prediction method undergoes 10 experimental runs, with the average results compared in Table 2.

2) SMALL SAMPLE PROMPT WORDS FOR EXTRACTING THE NUMBER OF AFFECTED PEOPLE

While current large language models demonstrate impressive capabilities in text generation and complex contextual understanding, they inherently lack deep comprehension of specific task requirements and rely heavily on explicit semantic guidance for effective output. This limitation can

TABLE 2. Comparison of error indicators of different prediction models.

Model	MSE	RMSE	MAE
WOA-BP	0.104	0.373	0.126
BP	0.145	0.402	0.152
LSTM	0.167	0.392	0.166
GM(1,1)	0.141	0.411	0.175

be addressed through prompt engineering, where structured prompting paradigms effectively bridge the gap between the model's general language understanding and the demands of domain-specific tasks—ensuring both output quality and task relevance.

This study enhances the application efficacy of general-purpose language models in vertical domains by constructing domain-adapted prompt templates. By aligning with task-specific needs and leveraging minimal annotated examples to activate the model's self-learning capability, this approach reduces reliance on specialized NLP models. Moreover, it significantly cuts computational and time costs for task-specific deployments by eliminating the need for large-scale fine-tuning. In disaster response scenarios—such as casualty statistics—this method proves particularly advantageous. When processing unstructured disaster data, the system rapidly constructs information extraction templates from limited labeled samples, dynamically adapting to evolving disaster conditions while maintaining accuracy.

The methodology for information extraction using large language models involves three steps:

Prompt Construction: For an input sentence X, design a prompt comprising task instructions, few-shot demonstrations, and the sentence itself. **Model Inference:** Feed the prompt to a large language model to generate a text sequence W. **Output Transformation:** Convert W into a sequence of entity tags for the final result. We evaluated three high-performance reasoning models extracting disaster-related figures. Below is an illustrative prompt template:

Task: Given the sentence:

“{Sentence}” related to an earthquake—analyze comprehensive contextual factors (geography, weather, socio-demographic impact, seasonal climate, etc.).

Target Entities: Affected population, deaths, injuries, precise location, city, time.

Identify which entities from the list appear in “{Sentence}”. For absent entities, infer plausible values based on disaster context and mark them with “[inferred]”.

Format Response As:

Given Sentence: {“BREAKING: A 6.1-magnitude earthquake struck Lushan County, Ya'an, killing 12, injuring 38, and displacing 14,928. 2013-04-20”}

Entities: {14,928, 12, 38, Lushan County, Ya'an City, 2013-04-20}

For missing data in example texts, populate columns with placeholder tags.

3) REASONING OF MATERIAL DEMANDS THROUGH THE THINKING CHAIN

The problem of predicting the types of materials and the per capita need for materials is considered step by step using the chain of thought technology. The thought chain is as follows:

User question: At 8:02 on April 20, 2013, a magnitude 7.0 earthquake occurred in Lushan County, Ya'an City, Sichuan Province (30.3 N latitude, 103.0 E longitude). The focal depth is 13 kilometers. Now I need to carry out an emergency rescue material supply plan. Please tell me which emergency rescue materials need to be prepared and what the per capita demand for each type of emergency rescue material is.

a: THE FIRST STEP IS TO ANALYZE THE DISASTER PARAMETERS

- Input: Basic seismic parameters such as magnitude, focal depth, and intensity distribution
- Processing: Query the emergency knowledge base and match the per capita material demand data

b: THE SECOND STEP IS TO CONSIDER THE INTEGRATION OF DISASTER INFORMATION

- Considering geographical spatial characteristics: Assessment of terrain structure vulnerability (mountainous areas/Plains/coastal areas)
- Consider population characteristics: population density and distribution of special groups in the affected areas
- Consider the infrastructure characteristics: the carrying capacity of the transportation network/medical institutions/shelters
- Consider dynamic variables: meteorological warning/probability of secondary disasters/Range of communication interruption
- Consider the emergency rescue stage: golden rescue period → transitional resettlement period → Reconstruction preparation period

Query the data of online resources and supplement and improve the disaster information

c: THE THIRD STEP IS TO CONSIDER THE LOGIC OF REQUIREMENT GENERATION

Based on the three stages of the golden rescue period (0-72 hours), the transitional resettlement period (72-240 hours), and the reconstruction preparation period (240+ hours), the prediction is dynamically adjusted to generate the types of required materials and the per capita material demand data. Based on the obtained casualty figures and the per capita material demand quantity, a material demand forecast is produced.

C. THE ANALYTIC HIERARCHY PROCESS (AHP) EVALUATES THE GENERATED CONTENT

Our testing utilized the April 20, 2013 M7.0 earthquake (8:02 AM Beijing Time, 30.3°N, 103.0°E focal coordinates) in Lushan County, Ya'an City, Sichuan Province as

a benchmark case. With a focal depth of 13km, this seismic event generated strong tremors across multiple regions including Ya'an, Chengdu, and Leshan in Sichuan Province, as well as Baoji, Hanzhong, and Ankang in Shaanxi Province. Official reports from Ya'an Emergency Management Office indicated catastrophic infrastructure damage in Longmen Township - over 99% building collapse, paralyzed medical services, and complete utilities shutdown, affecting 1.52 million residents across 12,500 km². The disaster resulted in 196 fatalities, 21 missing persons, and 11,470 injuries.

To address potential model overfitting risks from historical earthquake data inherent in reasoning-enhanced large language models' training corpus, we designed a simulated seismic scenario with distinct seasonal and geographic parameters not present in real-world precedents. Our simulated M6.5 earthquake scenario, hypothetically occurring in December 2025 at Lushan County, incorporated artificially generated information streams via temporal adjustments, manual simulations, and LLM-generated content, producing 2,000 multi-format digital records mimicking post-disaster communications (official bulletins, news reports, social media posts).

Given the absence of verified actual demand statistics for emergency supplies in major earthquakes, with available data predominantly reflecting reserve inventories rather than genuine post-disaster requirements, our evaluation framework employed expert validation. We convened 15 emergency management specialists from Ya'an City with over 5 years field experience, including veterans of the 2013 Lushan earthquake relief operations, to assess the model results. This deliberate selection of Ya'an as dual-scenario test location enhanced evaluation reliability through experts' contextual familiarity.

To ensure rigorous and objective model assessment, we implemented Analytic Hierarchy Process (AHP) analysis on expert ratings across four dimensions:

- **Comprehensibility:** Measuring semantic clarity and interpretability of predictions (higher = more understandable)
- **Accuracy:** Evaluating domain-specific knowledge correctness from professional databases and disaster context extraction (higher = greater precision)
- **Executability:** Assessing practical feasibility of proposed actions, including material categories and response protocols (higher = better implementability)
- **Comprehensiveness:** Benchmarking holistic consideration of disaster variables, e.g., seasonal hydration needs or alpine climate adaptations (higher = more thorough analysis)

This multidimensional evaluation framework enables systematic identification of optimal reasoning-enhanced LLM performance in predictive tasks through structured expert judgment quantification. In this paper, AHP is divided into three levels, including target level, first-level index, second-level index and third-level index. The evaluation framework for emergency response information quality

comprises four primary dimensions, each with specific sub-criteria. Comprehensibility assesses lexical complexity through sentence simplicity and common vocabulary usage, and logical coherence via proper connective usage and accurate key point identification. Accuracy evaluates source reliability by measuring authoritative citations and multi-source verification, along with data freshness through recent data ratios and historical revisions. Executability examines resource appropriateness considering material matching and local adaptation, and implementation capability regarding allocation specificity and resource availability. Comprehensiveness covers basic needs including environmental factors and demographic inclusion, plus special scenarios like extreme risks and chain reaction analysis. Each sub-criterion contains specific metrics, such as scoring higher for greater proportions of simple sentences or cross-verified information, ensuring systematic quality assessment across linguistic clarity, factual precision, operational feasibility, and holistic coverage.

The judgment matrix is a fundamental component of the Analytic Hierarchy Process (AHP). In the judgment matrix, elements at the same level need to be compared pairwise and assigned certain values. Generally, the nine scales from 1 to 9 are used as the judgment scales for the elements in the judgment matrix. Compare the importance of each two elements in the matrix one by one to obtain the judgment matrix $A_{n \times n}$ of order n .

$$A_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{1..} & a_{1n} \\ a_{21} & a_{22} & a_{2..} & a_{2n} \\ a_{..} & a_{..} & a_{..} & a_{..} \\ a_{n1} & a_{n2} & a_{n..} & a_{nn} \end{bmatrix} \quad (4)$$

The geometric mean method is used to divide m experts ($m=1,2,\dots$) The scoring matrix formed by k is multiplied bit by bit and then taken to the power of m to obtain the unique integrated matrix $\bar{A}$. In this study, $m=15$, and the formula is as follows:

$$\bar{A} = \left(\prod_{k=1}^m a_{ij}^k \right)^{\frac{1}{m}} \quad (5)$$

The weights of the integrated unique matrix are calculated by the geometric mean method (root square method), and the formula is as follows:

$$W_i = \frac{(\prod_{j=1}^n a_{ij})^{\frac{1}{n}}}{\sum_{i=1}^n (\prod_{j=1}^n a_{ij})^{\frac{1}{n}}} \quad i = 1, 2, 3 \dots n \quad (6)$$

Multiply the elements of $\bar{A}$ by rows to obtain a new vector. Then, take the power of n for each component of the new vector, and normalize the resulting vector to get the weight vector.

In practice, when experts compare the indicators pairwise, inconsistent conclusions may occur. Therefore, it is necessary to conduct consistency tests on the existing judgment matrices to ensure the rationality of the indicator weights. Generally, CR is taken as the criterion for judging the

TABLE 4. Weights of third-level indicators.

Level 1 Indicator	Relative Weight 1	Level 2 Indicator	Relative Weight 2	Level 3 Indicator	Relative Weight 3	Integrated Weight
Comprehensibility	0.1107	Lexical Complexity	0.3396	Sentence Length	0.2547	0.0096
				Common Vocabulary Usage	0.7453	0.0280
		Logical Coherence	0.6604	Logical Connectives	0.3443	0.0252
				Key Points Elaboration	0.6557	0.0479
Accuracy	0.3687	Source Reliability	0.7827	Authoritative Citations	0.6137	0.1771
				Information Verification	0.3863	0.1115
		Data Freshness	0.2173	Recent Data Ratio	0.3057	0.0245
				Historical Data Revision	0.6943	0.0556
Executability	0.3362	Resource Appropriateness	0.6332	Material Category Matching	0.6108	0.1300
				Local Condition Adaptation	0.3892	0.0829
		Implementation Capability	0.3668	Allocation Specificity	0.2672	0.0330
				Resource Accessibility	0.7328	0.0904
Comprehensiveness	0.1844	Basic Needs Coverage	0.6651	Environmental Factors	0.558	0.0684
				Demographic Coverage	0.442	0.0542
		Special Scenarios	0.3349	Extreme Scenarios	0.4425	0.0273
				Chain Reaction Analysis	0.5575	0.0344

TABLE 5. The calculation result of consistency check.

λ_{max}	CI	CR
4.0584	0.01948	0.0219<0.1

Among them, F_{LCS} represents the F1 score of ROUGE-L, and R_{LCS} represents the recall rate of ROUGE-L. P_{LCS} represents the ROUGE-L precision.

The evaluation of diversity is used to measure the types and breadth of information retrieved and generated by RAG. Its purpose is to ensure that the system can provide diverse perspectives and avoid the occurrence of response redundancy. RAG will evaluate diversity by calculating the cosine similarity between different retrieval results or generated texts. When the cosine similarity score is low, This means higher diversity, which indicates that RAG can provide diverse responses from a wide range of information sources. The calculation formula of cosine similarity is as follows:

$$\cos \theta = \frac{A \cdot B}{||A|| \cdot ||B||} \quad (14)$$

Here, $A \cdot B$ represents the dot product product of vector A and vector B. $||A||$ and $||B||$ respectively represent the modules of vectors A and B, and θ represents the Angle between the two vectors.

Experimental results demonstrate that the integration of retrieval-augmented generation has improved response

TABLE 6. Model retrieval generates effect evaluation.

Model	ROUGE-L	MRR	Cosine similarity
DeepSeek-R1	0.893	0.833	0.819
GPT-o3-mini	0.821	0.847	0.802
Qwen/QwQ-32B	0.784	0.856	0.795
DeepSeek-R1 without RAG	0.798	/	0.624
GPT-o3-mini without RAG	0.736	/	0.729
Qwen/QwQ-32B without RAG	0.665	/	0.828

quality across all three models while exhibiting superior diversity compared to non-RAG large language models. Since non-RAG models were not subjected to retrieval effectiveness testing, their MRR values remain unrecorded. The DeepSeek-R1 model demonstrates commendable performance in text generation quality and semantic alignment, as evidenced by its ROUGE-L score of 0.893 and cosine similarity of 0.819. However, its relatively low MRR score suggests a higher propensity for hallucination phenomena in generated responses. The Qwen/QwQ-32B model achieves the highest MRR score, indicating superior performance in information

localization tasks, while maintaining balanced performance across all three metrics consistently above 0.75, illustrating stable comprehensive capabilities.

Different models exhibit distinctive strengths in text generation, semantic comprehension, and task prioritization, necessitating model selection based on specific application requirements. For scenarios prioritizing generation quality, DeepSeek-R1 is recommended. GPT-o3-mini proves more suitable for retrieval-ranking emphasis, while Qwen/QwQ-32B represents an optimal choice for applications requiring balanced multi-dimensional performance, particularly recommended for offline deployment contexts.

B. EVALUATION OF THE EFFECT OF MATERIAL DEMAND FORECASTING

We used three large language models to conduct two predictions for two earthquakes respectively, obtaining 12 prediction results, which were provided to experts for scoring, and a total of 180 scoring results were obtained. We simulated an earthquake under different seasonal and geographical conditions that did not actually occur, and used the model to predict the material demand to avoid the influence of the historical data that the basic reasoning large model has learned. The scoring results are shown in Figure 4 and Figure 5.

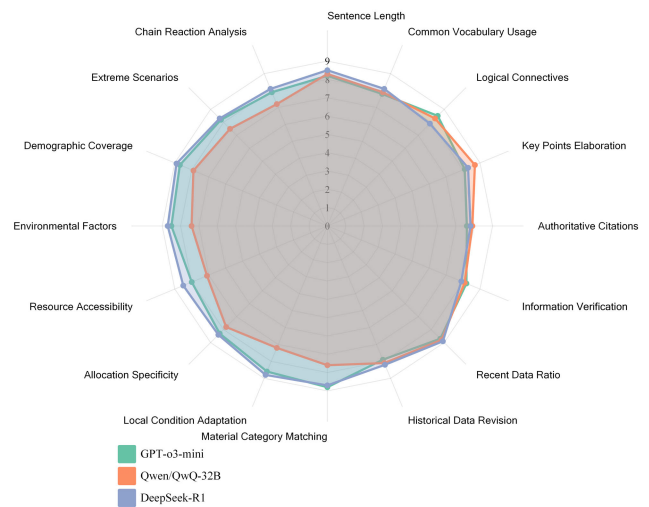


FIGURE 4. Evaluation of content generated regarding the Magnitude 7.0 Earthquake in Lushan county, Ya'an city on April 20, 2013.

It can be seen from Table 7 for the magnitude 7.0 earthquake in Lushan County, Ya'an City on April 20, 2013, responses generated by three large language models demonstrated commendable quality, achieving weighted average scores above 7. This indicates that our web-based emergency rescue material demand prediction methodology can essentially produce forecasts meeting emergency responders' expectations in historical cases.

Regarding the simulated magnitude 6.5 earthquake in Lushan County, Ya'an City on December 1, 2025, responses generated by the three large language models exhibited

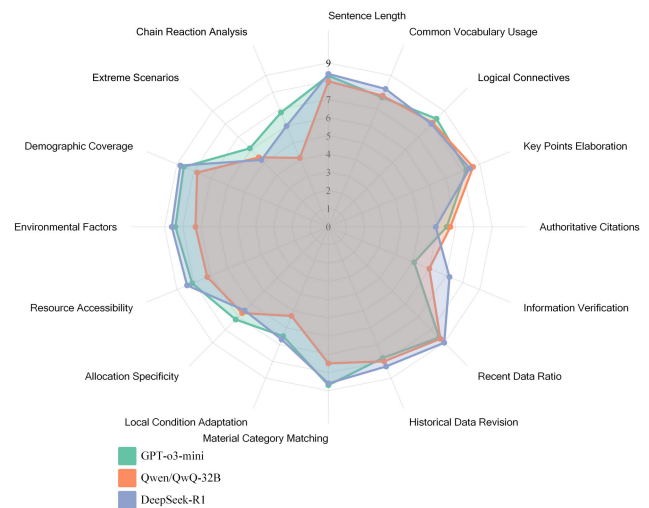


FIGURE 5. Evaluation of content generated regarding the simulated magnitude 6.5 Earthquake in Lushan county, Ya'an city on December 1, 2025.

TABLE 7. The weighted average score of the indicator.

Model	Earthquake in Lushan County, Ya'an City on April 20, 2013	Earthquake in Lushan County, Ya'an City on December 1, 2025
GPT-o3-mini	8.21456	7.34474
Qwen/QwQ-32B	7.76586	6.92167
DeepSeek-R1	8.33006	7.50472

declines across multiple metrics, with weighted average scores dropping to 6–7. Primary decreases occurred in indicators requiring professional expertise and web-based reasoning analysis, such as proportion of authoritative citations, contingency planning for extreme scenarios, and cascading effect analysis. This deterioration may be attributed to insufficient volume of simulated web-based resource data.

Nevertheless, DeepSeek-R1 and GPT-o3-mini maintained their performance in comprehensiveness- and clarity-related metrics, underscoring both the fundamental capabilities of large language models and the practical utility of employing them for holistic disaster assessment.

C. APPLICATION CASE

This material prediction scheme was applied and tested using the data of the 6.8-magnitude earthquake in Dingri County, Xigaze City, Tibet, China in 2025.

The earthquake occurred at 9:05 a.m. on January 7, 2025, Beijing time. The epicenter was located in Cuoguo Township, Dingri County, Xigaze City, Tibet (28.50 degrees north latitude, 87.45 degrees east longitude), and the focal depth was 10 kilometers. After the earthquake occurred, we immediately began to crawl relevant online resource information and successively carried out work such as information classification, obtaining the number of affected

people after the earthquake, and dynamic prediction of material demands.

Before the launch of Deepseek-R1, the entire workflow was completed using ChatGPT. However, when the non-reasoning large language model predicted the types and quantities of materials, its interpretability was poor due to the lack of demonstration of the thinking process. After passing the experimental tests and expert scoring in the previous chapter section, Deepseek-R1 was reused as the basic model to conduct application tests on the earthquake case data, and good application results were achieved.

After the earthquake occurred, we crawled information related to the earthquake, such as government announcements, news articles, and social media comments. The crawling operation lasted for a total of 15 days until the level one emergency response to the earthquake was terminated on January 21st. A total of 14,372 pieces of data were crawled using the keyword “fixed date earthquake”. After cleaning and deduplication, 7,207 pieces remained. Among them, social media and self-media data accounted for 6,988 pieces, media report data 117 pieces, and government announcements and articles 102 pieces.

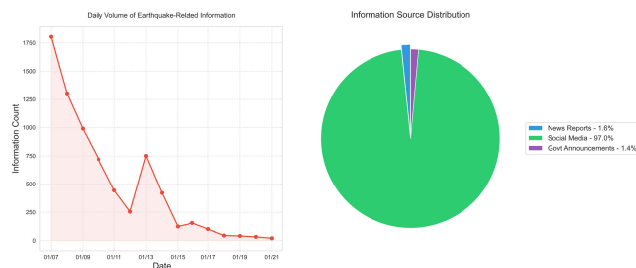


FIGURE 6. The 6.8-magnitude earthquake that struck Dingri County, Xigaze City, Tibet, China in 2025.

From the Figure 6, we can observe that on the first day after the earthquake, there was the most related information and the highest public opinion heat on the Internet, and then it gradually decreased. On January 13th, the public opinion heat rose again because the largest aftershock after the earthquake occurred that day, which once again sparked heated discussions online.

Based on emergency knowledge databases and online information retrieval-augmented generation for large language models, the following inquiry is proposed:

An earthquake occurred at 09:05 Beijing time on January 7, 2025, with its epicenter located in Coguo Township, Tingri County, Shigatse City, Tibet (28.50 °N, 87.45 °E), featuring a focal depth of 10 kilometers. Please specify the appropriate categories of emergency supplies required and the recommended per capita quantities for each category. Ensure comprehensive consideration of temporal parameters (seasonality), geographical and topographical conditions, socioeconomic circumstances in impacted areas, and cultural sensitivities in local communities.

The thinking process of the large model is as follows:

Ok. The user needs a detailed form on the preparation of earthquake emergency materials in Dingri County, Tibet. Factors such as time (winter in January), geographical location (high altitude, remote), social economy (possible difficulties in material transportation), and humanistic environment (Tibetan-inhabited areas) need to be taken into consideration. I need to determine the influence of these factors on the demand for materials first.

First of all, it's January. Winter in Tibet is extremely cold and the temperature may be very low. Therefore, warm supplies are the top priority, such as down jackets, sleeping bags and heating equipment. Meanwhile, traffic may be disrupted due to snowfall or road damage, and the reserve quantity and transportation routes of emergency supplies need to be considered.

In terms of geographical environment, Dingri County is a high-altitude area with an average altitude of over 4,000 meters. The air is thin and there may be a risk of altitude sickness. Therefore, it is necessary to prepare relevant medicines such as rhodiola rosea and oxygen cylinders. In addition, the local area may mainly rely on animal husbandry, and the food supply may be insufficient. It is necessary to consider high-calorie and long-lasting foods, such as compressed biscuits and instant noodles, as well as sufficient drinking water.

In terms of economic and social factors, Xigaze City is a relatively remote area, and its medical resources may be limited. It is necessary to prepare sufficient medical first aid kits and basic medicines. At the same time, considering the eating habits of the Tibetan people, it may be necessary to prepare traditional foods such as butter tea and Tsampa to meet the local people's dietary needs and to avoid other problems caused by food maladjustment.

In terms of the humanistic environment, respecting local culture and language communication is very important. Bilingual (Chinese and Tibetan) rescue guidelines or manuals may be needed to ensure that information is conveyed accurately. In addition, considering that some temples or religious sites may be damaged, the rescue supplies should include necessary religious items for support. However, users may not have mentioned this point, so it can be temporarily disregarded unless there is a clear need.

It is also necessary to consider the damage to infrastructure caused by the earthquake, such as power outages. Therefore, emergency lighting equipment such as flashlights, headlamps and spare batteries are very important. For temporary accommodation, tents or disaster relief tents may be needed. Considering the high-altitude and cold regions, the tents need to have good heat insulation performance.

Then, the per capita quantity needs to be estimated reasonably. For instance, each person needs at least 2 liters of water per day. However, considering the transportation issue, it might take about 3 days initially, so the average person needs 6 liters. In terms of food, for the same three days of high-calorie food, for instance, 2,000 kilocalories are

TABLE 8. Emergency material standards for high-altitude disasters.

Category	Material Name	Per Capita Quantity	Key Specifications
Warm Protection	Thick waterproof down jacket	1 piece	Night temperature drops to -20°C in Tibet during January. Requires windproof/waterproof design, preferably adapted from traditional Tibetan robes.
	Alpine sleeping bag (-25°C)	1 unit	Must include moisture-proof pad, specially designed for damp high-altitude terrain.
	Self-heating warm patches (18h)	10 patches/3 days	3-4 patches/day, requires instructions in Tibetan.
Survival Supplies	Emergency food (500kcal/pack)	9 packs/3 days	Daily ration of 3 packs, prioritizing high-calorie Tibetan-style modified food containing butter and tsampa.
	Portable oxygen tank (10L)	0.5 tank	1 tank shared between 2 people, meeting basic needs for 72 hours.
	Drinking water (1.2L)	1 bottle/day	Includes water purification tablets (6 tablets/person) for melting snow water.
Medical Support	High-altitude first aid kit	0.2 kit	Contains altitude sickness medication (6 Rhodiola capsules), frostbite ointment, pulse oximeter, Tibetan-Chinese bilingual first aid guide.
	Pulse oximeter	1 unit/family	Must function properly in -20°C environment for altitude sickness monitoring.
Shelter	Portable tent	1 tent/5 people	Made of windproof insulating materials to meet thermal requirements.

needed each day. Compressed biscuits have approximately 500 kilocalories per pack, so 4 packs a day, totaling 12 packs over three days. However, such a calculation might be too detailed. Users might need more simplified suggestions, such as 3 packs per person per day, totaling 9 packs for three days.

In terms of warmth, each person gets a down jacket, a thickened sleeping bag, and a blanket, but a sleeping bag might be more practical. Heating items like heat packs may require 4 to 5 pieces a day, approximately 15 pieces for three days. However, considering actual usage, 10 pieces might be sufficient.

In terms of medical care, it might be more appropriate for each family to have one first aid kit. For each person, one is fine. Prepare three days' worth of altitude sickness medication for each person. Oxygen cylinders may need to be shared, so 0.5 cylinders per person? Or it can be provided by family, but if the user requests per person, a specific value may be needed. For example, if there is one bottle for every two people, then 0.5 per person. However, in practice, it may be more inclined towards the basic needs of each person, and the standards need to be further confirmed.

Other items such as tools may require one safety helmet and one multi-functional shovel for each family group. However, if the user requests per person, adjustments might be needed here. For instance, one safety helmet per person

and one shovel for every five people. But in the table, it might be necessary to uniformly calculate per person. This part needs to refer to the standards of emergency supplies.

Finally, considering the difficulties in transportation, the materials need to be light and portable, and at the same time, a combination of local government reserves and external assistance may be required. It is necessary to ensure that the types of materials are comprehensive, the quantities are reasonable, and they are suitable for the local environment. Now these factors need to be integrated into the table by type, material name, per capita quantity, description, and ensure that each category covers different demands.

After manually organizing the answers, the prediction results of the large model for the types of materials and per capita demand are as follows Table 8.

Compared with the traditional prediction schemes as shown in the Table 9, this scheme can make more refined considerations for the materials needed in earthquakes. It comprehensively considers factors such as the season, geographical environment, and ethnic culture of the occurrence of deterministic earthquakes, and provides more types of material predictions. For common emergency rescue supplies such as drinking water, clothing and tents, remarks and descriptions that are more in line with the actual needs of emergency rescue in Tibetan areas have also been provided.

TABLE 9. Comparison of material demand types.

Material Types	Ref.1 [15]	Ref.2 [21]	Ref.3 [36]	Research Proposal
Drinking water	✓	✓	✓	✓
Clothing	✓	✓	×	✓
Food	✓	×	✓	✓
Medical supplies	✓	×	✓	✓
Tents	✓	×	✓	✓
Quilts	×	×	✓	✓
Emergency lighting equipment	×	×	×	✓
Oxygen cylinders	×	×	×	✓
Self-heating warm patches	×	×	×	✓
Bilingual rescue manual (Chinese-Tibetan)	×	×	×	✓
Pulse oximeter	×	×	×	✓

After the earthquake occurred, we immediately made a preliminary prediction of the possible number of people affected by the earthquake based on factors such as the magnitude of the earthquake, the depth of the epicenter, the longitude and latitude of the epicenter location, the population density of the epicenter area, and the time of the earthquake. Meanwhile, the number of affected people is extracted from the crawled network resources to dynamically correct the number of affected people.

In the early post-earthquake period, the predicted number of affected people by WOA-BP was 76,300. On the evening of January 7th, 30,400 affected people were extracted. On January 9th, the official news released that the number of affected people was 61,500. We promptly corrected the number of affected people. The important information extracted is as follows:

- At 11:00 on January 7, 2025, nine people were killed.
- At 12:00 on January 7, 2025, the earthquake had claimed 32 lives and injured 38 people to varying degrees.
- At 12:30 on January 7, 2025, the earthquake in Dingri County, Xigaze City, has caused 53 deaths and 62 injuries.
- At 15:00 on January 7, 2025, the earthquake had caused 95 deaths and 130 injuries.
- On the evening of January 7, 2025, over 12,000 people from various levels of emergency rescue, fire rescue, forest fire rescue, public security officers, military personnel and social emergency rescue forces carried

- out a full-scale search and rescue operation. They rescued 407 trapped people and set up 14 transfer and resettlement sites to transfer and resettle over 30,400 affected people. With the support of the rescue forces, some resettlement sites have begun to supply hot meals.
- At about 11:00 on January 8, 2025, a total of 14,668 people from the military, public security, armed police, fire department, forest fire department, border management, as well as Party members, cadres and the public were coordinated and mobilized to participate in the earthquake relief work. A total of 407 trapped people were rescued. A total of 26 medical rescue teams with 581 personnel and 107 ambulances were dispatched at the district, municipal and county levels to carry out the transfer and treatment work. All the injured were admitted to the hospital, and one seriously injured person was transferred to the autonomous regional-level hospital for treatment.
- Sufficient forces have been mobilized to fully carry out the restoration and repair work of damaged roads, power supply, communication and other facilities. All the sections damaged by the earthquake have been restored, and the mobile networks and power supply in the three towns of Changsuo, Cuoguo and Quluo, the epicenter of the earthquake, have been fully restored.
- We have made every effort to carry out the resettlement work for the affected people. A total of over 428,000 items (pieces) of essential supplies such as cotton tents, folding beds, bunk beds, cotton quilts, blankets, cotton overcoats (cold-proof suits), cold-proof shoes, heaters, generators, stoves, emergency lighting lamps, as well as food and medicine, and 81 truckloads of fuel have been transported to the disaster-stricken areas one after another.
- A total of 187 resettlement sites were set up, with 3,705 tents erected and 46,525 people relocated and resettled. The basic living standards of the affected people were guaranteed.
- On the afternoon of January 8, 2025, the official account of Dingri County Media Convergence Center, “Window of Mount Everest”, released an announcement from Dingri County regarding the cessation of receiving disaster relief donation materials.
 - As of 6 a.m. on January 9, 2025, the earthquake has claimed 126 lives, affected approximately 61,500 people to varying degrees, set up 224 resettlement sites, and relocated 47,500 people. The earthquake has also damaged 27,248 households, among which 3,612 collapsed.
- Based on the dynamically obtained network information, we continuously revise the initial prediction results to achieve the dynamic prediction of emergency material demands. According to the information on the extracted material donations, even if calculated based on the predicted maximum number of 76,300 people in the early post-earthquake period, excluding special items such as blood oxygen meters, the

TABLE 10. Emergency material demand estimation.

Category	Material Name	Per Capita Quantity	Initial Post-disaster Estimate	Revised Demand (Jan 7)	Revised Demand (Jan 9)
Warmth Protection	Thick water-proof down jacket	1 piece	76,300 pieces	30,400 pieces	61,500 pieces
	High-altitude sleeping bag (-25°C)	1 unit	76,300 sets	30,400 sets	61,500 sets
	Self-heating warm patch (18h)	10 pieces/3 days	763,000 pieces	304,000 pieces	615,000 pieces
Survival Supplies	Emergency food (500 kcal/pack)	9 packs/3 days	686,700 packs	273,600 packs	553,500 packs
	Portable oxygen tank (10L)	0.5 tank	38,150 tanks	15,200 tanks	30,750 tanks
	Drinking water (1.2L)	1 bottle/day	228,900 bottles	91,200 bottles	184,500 bottles
Medical Support	High-altitude first aid kit	0.2 set	15,260 sets	6,080 sets	12,300 sets
	Oximeter	1 unit/family	15,260 units	6,080 units	27,248 units
Shelter	Portable tent	1 tent/5 people	15,260 tents	6,080 tents	27,248 tents

donated and allocated materials from January 7th to January 8th alone have fully met the emergency material demands. On January 8th, the official of Dingri County released information to stop accepting material donations. This is consistent with the result of our speculation based on the predicted quantity and the quantity of extracted materials, indicating the practicability of this research plan.

V. CONCLUSION

Earthquakes, characterized by their sudden onset and high destructiveness, pose a significant threat to the safety of people's lives and property. The rational supply of emergency relief materials post-earthquake is of critical importance, and accurate demand prediction serves as the cornerstone for achieving this rational allocation. Traditional prediction methods face multiple limitations when addressing the complex and dynamic scenarios of earthquake disasters. On one hand, they struggle to adapt flexibly to varying conditions across different disaster zones—such as the

pronounced differences between mountainous regions and plains. On the other hand, evolving disaster situations, including fluctuating casualty counts or emergent secondary disasters like landslides, necessitate real-time adjustments to demand forecasts.

To address these challenges, this study leverages reasoning-enhanced large language models (LLMs) integrated with retrieval-augmented generation (RAG) technology to construct a dynamic prediction framework for post-earthquake emergency material demand. By synthesizing online information with emergency knowledge bases, the framework achieves differentiated and adaptive demand forecasting. Key research contributions are outlined below:

A. DISASTER FEATURE EXTRACTION FROM WEB RESOURCES

Employing natural language processing (NLP) techniques, critical disaster characteristics such as earthquake magnitude, terrain type, population density, and economic conditions are automatically extracted from news reports, social media, and geospatial datasets. These granular features enable differentiated demand analysis, overcoming the inflexibility of traditional methods to rapidly assess disparities between disaster zones.

B. CONTEXT-AWARE DEMAND PREDICTION

Guided by chain-of-thought (CoT) prompting, the model emulates human-like stepwise reasoning—for instance, predicting medical supplies by first analyzing injury patterns (e.g., fractures or crush injuries) before mapping them to specific pharmaceuticals and equipment. In scenarios like winter earthquakes in remote mountainous areas, the system identifies contextual features (e.g., subzero temperatures and transportation bottlenecks) to prioritize forecasts for insulation materials and logistics support, ensuring logical and scenario-appropriate outputs.

An integrated emergency knowledge base, comprising national material standards, historical earthquake responses, and rescue protocols, further enhances predictions. For example, the model references benchmark data (e.g., first-aid kits per 1,000 affected individuals) from similar past disasters. Continuous updates from real-time web sources—such as weather fluctuations or road repair progress—enable dynamic knowledge refinement, while RAG minimizes LLM hallucinations and reinforces output reliability.

C. REAL-TIME ADAPTIVE FORECASTING

A key challenge in demand prediction lies in the discrepancy between available data (e.g., warehouse inventories and donation records) and actual needs. To address this, the framework translates material gaps into population impact projections, combining per-capita demand coefficients with adjusted casualty estimates.

Using LLMs to analyze live field reports and responder feedback, the system dynamically recalibrates predictions for both affected populations and material categories

(e.g., sudden spikes in shelter needs due to aftershocks). This iterative approach ensures alignment with on-ground realities, optimizing resource allocation efficiency during critical relief operations.

D. FUTURE PERSPECTIVES

While this study represents preliminary progress in emergency supply forecasting, further research remains essential to address the dynamic complexities of earthquake disasters and better support relief operations. Compared to well-established linguistic datasets in fields like medicine and law, disaster research still lacks high-quality Chinese natural language processing corpora. Building a comprehensive, high-quality Chinese disaster information dataset should be prioritized.

Currently, the framework predominantly utilizes textual data and natural language processing. However, online resources also contain valuable visual, auditory, and sensor-based data (e.g., images, videos, seismic monitoring feeds). Extending the model to a multimodal architecture—integrating IoT devices such as surveillance cameras and seismometers—could significantly enhance predictive performance.

While initially designed for earthquakes, the framework can be adapted to other disasters (e.g., floods, typhoons) by modifying knowledge bases and algorithmic components. For instance, in flood scenarios, predictions could incorporate boats and life jackets, while typhoon preparedness might prioritize windproof and waterproof supplies. Such expansions would substantially broaden the model's applicability.

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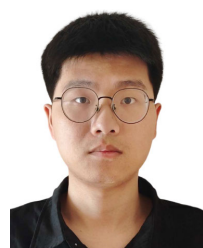


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